Ca^{2+}tchAll

Calcium Imaging Data Analysis & Visualization

A browser-based superplot tool for multi-replicate calcium imaging experiments — upload, assign, and visualize in seconds.

USER MANUAL · v1.0.0-alpha · 2026

© 2026 Tomas Luyten · MIT License

Table of Contents

1 **Introduction** 3

What is Ca^{2+}tchAll? 3

Application Layout 3

Getting Started 4

2 **Loading Files** 5

Supported File Format 5

Upload Workflow 5

Assigning Conditions 6

Session Memory Bar 7

Upload Limits & Errors 7

3 **Metric Tabs** 8

Standard Metrics 8

TG Metrics 9

Add-Back Metrics 9

Trace Tabs (Raw F & $\Delta F/F_o$) 9

4 **Chart Styles** 10

5 **Plot Controls** 11

Summary Statistics 11

Error Bars 11

Data Representation 12

Condition Ordering 12

Point Style Sliders 12

6 Appearance Panel 13
Condition Colors 13
Labels & Title 14
Typography 14
Axis Ranges 14
7 Exporting Plots 15
8 Preferences & Persistence 15
9 Onboarding Tour 16
10 Dark / Light Theme 16
11 Keyboard Shortcuts 16
12 Troubleshooting 17

SECTION 1

Introduction

What Ca^{2+}tchAll does and how the interface is organized

1.1 What is Ca^{2+}tchAll?

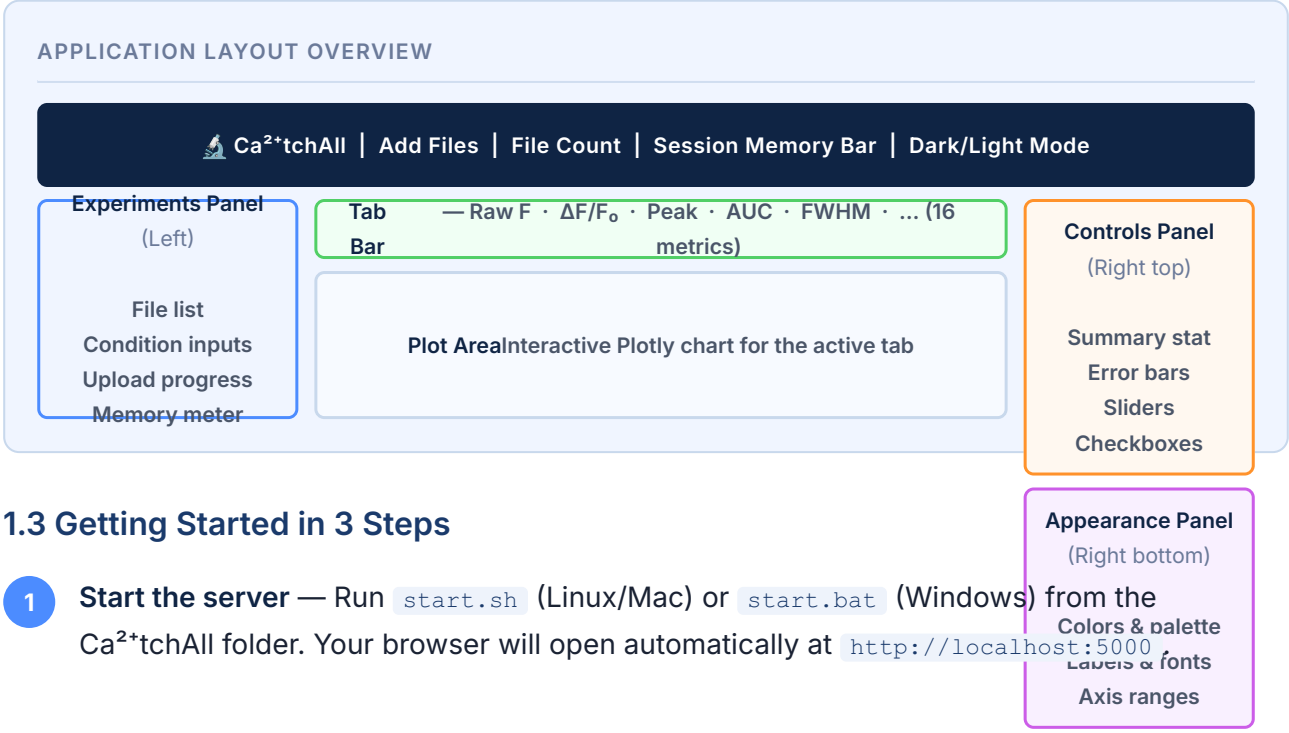
Ca^{2+}tchAll is a browser-based analysis and visualization tool for multi-replicate calcium imaging experiments. It takes **.xlsx files exported from the Ca^{2+}tch-One** and instantly produces publication-quality superplots — charts that overlay individual replicate distributions alongside summary statistics.

The workflow is straightforward: upload your files, type a condition name next to each one, and the plots appear automatically. Everything runs locally on your computer — no data is ever sent to an external server.

> **i WHAT IS A SUPERPLOT?**
>
> A superplot shows both the underlying data distribution and the replicate-level summary in the same chart. Individual ROI values are shown as small dots; per-experiment means are shown as larger dots; the overall condition mean ± error bar is drawn on top. This is the format recommended by Lord et al. (2020) for transparent reporting of biological replicates.

1.2 Application Layout

The interface is divided into four main areas:



1.3 Getting Started in 3 Steps

1. **Start the server** — Run `start.sh` (Linux/Mac) or `start.bat` (Windows) from the Ca^{2+}tchAll folder. Your browser will open automatically at `http://localhost:5000`.

2. **Upload your files** — Click **Add Files** in the top header, or drag-and-drop `.xlsx` / `.xls` files onto the Experiments panel. Wait for the upload progress indicators to show "completed".

3. **Assign conditions** — Type a condition name (e.g. "Control", "Treated") next to each file. Files sharing the same condition name are grouped as biological replicates. Plots appear automatically as soon as at least one condition is assigned.

> 💡 **FIRST-TIME TIP**
>
> On your first visit, an interactive 5-step onboarding tour launches automatically. You can skip it at any time and replay it later by clicking the **?** button in the bottom-left corner.

SECTION 2

Loading Files

Uploading experiments, assigning conditions, and managing session memory

2.1 Supported File Format

Ca^{2+}tchAll reads **.xlsx** and **.xls** workbooks exported by the Ca^{2+}tch-One. The workbook must contain at least one of the following sheets:

Sheet name	Required columns	Content
Metrics	`roi_id` + metric columns	One row per ROI — scalar values for each measured metric
Raw_Traces	`Time_s` + `ROI_*` columns	Raw fluorescence intensity time series per ROI
DeltaF	`Time_s` + `ROI_*` columns	Normalized $\Delta F/F_0$ time series per ROI
Metadata (optional)	Key–value pairs	Experiment metadata (signal mode, etc.)
Settings (optional)	Key–value pairs	Acquisition settings (analysis mode, etc.)

⚠ TRACE SHEET SIZE LIMIT

Trace sheets with more than 500,000 cells (frames × ROIs) are automatically skipped with a warning to prevent browser slowdown. The Metrics sheet is always read in full regardless of size.

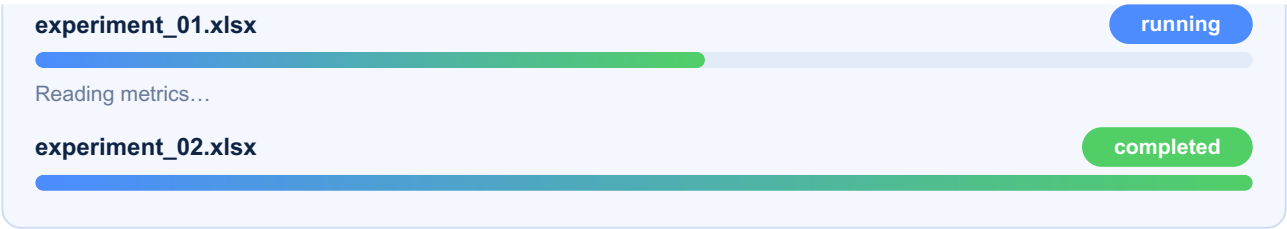
2.2 Upload Workflow

There are two ways to add files:

- **Click "Add Files"** in the top header bar — opens your operating system file picker. You can select multiple files at once.
- **Drag and drop** — drag one or more files from your file manager directly onto the Experiments panel on the left. The panel highlights with a blue border when a drag is detected.

Each file goes through the following stages shown in the upload job list:

UPLOAD JOB INDICATOR

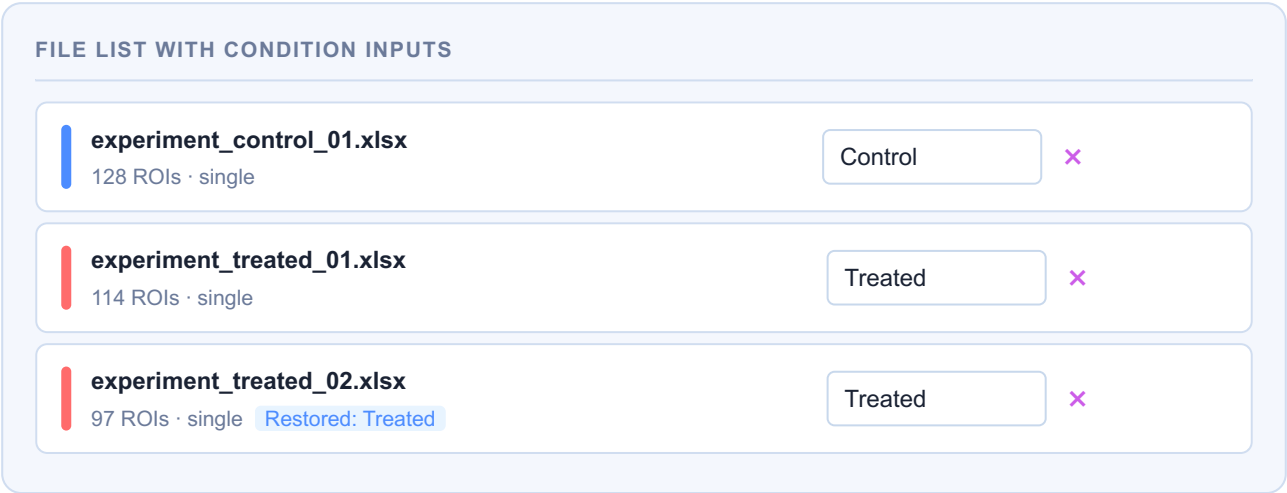


Stage	Meaning
queued	File received, waiting to be processed
running	Parsing in progress (validates structure, reads sheets, estimates memory)
completed	File loaded successfully — appears in the file list below
failed	File could not be loaded — see the error message for details
canceled	Upload was canceled by clicking the Cancel button during processing

Completed upload jobs can be dismissed by clicking **Dismiss** on each row. They disappear automatically when you reload the page.

2.3 Assigning Conditions

Once a file is loaded, it appears in the file list with a text input for the condition name:

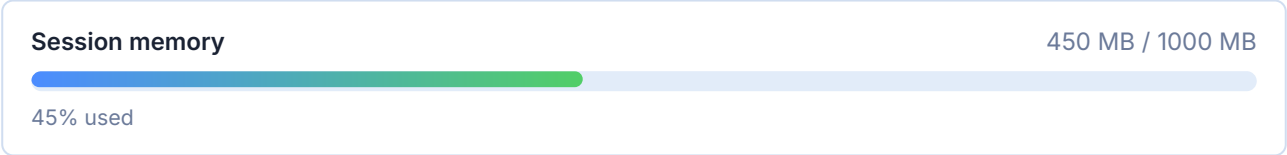


Key points about condition assignment:

- **Same name = same condition:** Files sharing an identical condition name are grouped as biological replicates of that condition (e.g. three files labeled "Control" form one condition with three replicates).
- **Autocomplete:** As you type, existing condition names appear as suggestions in a dropdown — click to apply.

- **Auto-restore:** If you have uploaded the same filenames before and assigned conditions, those assignments are restored automatically on re-upload. A Restored: [name] badge is shown.
- **Plots update instantly:** Each time you change a condition name, all plots refresh within a fraction of a second.
- **Remove a file:** Click the × button on any file row to remove it from the session immediately.

2.4 Session Memory Bar



The session memory bar in the top header tracks the total in-memory size of all loaded files. It updates after every upload or deletion.

- **Normal (0–70%):** All good — continue loading files.
- **Warning (70–90%):** The bar turns orange. Consider removing files you no longer need before adding more.
- **Danger (90%+):** The bar turns red. New uploads will be blocked if they would exceed the memory cap.

The memory cap is automatically calculated as 25% of your system's total RAM (minimum 512 MB), ensuring the app never destabilizes your machine.

2.5 Upload Limits & Error Messages

Limit	Default value	What happens when exceeded
Max files per session	Dynamic (RAM-based)	Upload blocked with "file-count limit" message
Per-file RAM	Up to 50% of session cap	Upload blocked with "per-file RAM cap" message
Session RAM total	25% of system RAM	Upload blocked with "total RAM cap" message
ROIs per file	4,000	Upload blocked with "ROI limit" message
Parse timeout	45 seconds	Upload fails with "parse timeout" message
Trace sheet size	500,000 cells	Trace sheet skipped, scalar metrics still loaded

SECTION 3

Metric Tabs

Understanding each calcium imaging metric and the trace visualization tabs

The tab bar at the top of the plot area shows one tab per available metric. Tabs are shown only for metrics present in the loaded files — if no file contains TG data, for example, those tabs remain hidden. The active tab is highlighted and its plot is shown in the main area.

3.1 Standard Metrics

These metrics describe individual calcium events detected in the recording:

Tab	Full name	Unit	What it measures
Peak	Peak amplitude	$\Delta F/F_o$	The maximum height of a calcium transient above baseline. A larger peak means a stronger calcium response.
AUC	Area Under the Curve	$\Delta F/F_o \cdot s$	The total calcium flux over the duration of an event (integral of the signal). Reflects both amplitude and duration combined.
FWHM	Event Full Width at Half Maximum	s	How long an event lasts measured at 50% of its peak amplitude. A longer FWHM means the calcium signal persists longer.
Frequency	Event frequency	Hz	How many calcium events occur per second in a cell. Higher frequency indicates more active calcium signaling.
TTP	Time to peak	s	How long it takes from event onset to reach the maximum amplitude. Reflects how fast the response rises.
Decay	Decay t½	s	The half-life of the falling phase — how long it takes the signal to return to 50% of peak after the maximum. Shorter means faster clearance.
Rise	Rate of rise	$\Delta F/F_o / s$	The slope of the rising phase of the calcium transient. A steeper rate means the signal climbs faster.

3.2 TG Metrics (Thapsigargin Protocol)

These metrics are specific to experimental protocols where thapsigargin (TG) is applied to deplete ER calcium stores. They only appear if your files contain TG response data.

Tab	Full name	Unit	What it measures

TG Peak	TG peak amplitude	$\Delta F/F_o$	Maximum calcium release triggered by ER store depletion
TG Slope	TG initial slope	$\Delta F/F_o$ / s	Steepness of the initial TG-induced calcium rise
TG AUC	TG area under curve	$\Delta F/F_o \cdot s$	Total calcium flux during the TG response

3.3 Add-Back Metrics

These metrics measure the calcium entry that occurs when extracellular calcium is re-added after TG-induced ER depletion (store-operated calcium entry, SOCE).

Tab	Full name	Unit	What it measures
AB Peak	Add-back peak	$\Delta F/F_o$	Peak calcium entry amplitude during add-back phase
AB Slope	Add-back initial slope	$\Delta F/F_o$ / s	Rate of calcium entry at the start of add-back
AB AUC	Add-back area under curve	$\Delta F/F_o \cdot s$	Total calcium entry during add-back response
AB TTP	Add-back time to peak	s	Time from add-back until peak calcium entry is reached

3.4 Trace Tabs (Raw F and $\Delta F/F_o$)

The first two tabs show **time-series traces** rather than scalar summary statistics:

- **Raw F:** Shows absolute fluorescence intensity over time for every ROI in every file. Traces are grouped and color-coded by condition, with individual replicates overlaid.
- **$\Delta F/F_o$:** Shows the normalized fluorescence change relative to the resting baseline. This is the standard way to compare calcium signals across cells and experiments independent of baseline fluorescence levels.

For trace tabs, the X axis range control (in the Appearance panel) allows you to zoom into a specific time window.

> **i TRACE AVAILABILITY**
>
> Trace tabs are only shown if the uploaded files contain **Raw_Traces** or **DeltaF** sheets. Files exceeding the 500,000-cell trace limit are noted as "traces skipped" in the file list but their Metrics data is still available.

SECTION 4

Chart Styles

Five visualization types for metric tabs — choose the one that best tells your data's story

The chart style selector is located directly below the tab bar. Changing style updates all metric plot tabs instantly. Trace tabs always show overlaid time-series regardless of the selected style.

CHART STYLE SELECTOR

Violin | Bar | Box | Strip | Raincloud

Style	What it shows	Best used when
Violin	A mirrored kernel density estimate showing the full probability distribution of values within each condition. Individual ROI points are overlaid as a scatter plot. Per-file means are shown as larger dots. The overall condition mean ± error bar is drawn on top of the violin.	You want to show the full data distribution and identify skewness, bimodality, or outliers. This is the default and recommended style for most calcium imaging data.
Bar	A classic bar chart showing the condition mean as bar height, with error bars (SEM, SD, or 95% CI) drawn on top. Individual ROI points are overlaid.	You need a compact, traditional visualization for a narrow figure panel, or when reviewers expect bar charts.
Box	A box-and-whisker plot showing the median (center line), interquartile range (box), Tukey whiskers (1.5× IQR), and outlier points beyond the whiskers.	You want to highlight the median and spread without assuming a normal distribution, or when comparing skewed datasets.
Strip	All individual data points shown as a vertical scatter plot, jittered horizontally to reduce overplotting. Per-file means are shown as larger dots. No distributional shape is estimated.	You have small sample sizes where individual points are meaningful, or you want the most transparent data display.
Raincloud	A "raincloud" combines a half-violin (density estimate) above with a strip plot of individual points below. Per-file means are overlaid. This style maximizes information density.	You want both the distributional shape and every individual point visible — ideal for publication figures.

💡 **WHICH STYLE TO CHOOSE?**

For publication figures, **Raincloud** or **Violin** are recommended as they show the full data distribution alongside the summary statistics, satisfying transparency requirements from most journals. **Bar** charts alone are considered insufficient by many reviewers.

SECTION 5

Plot Controls

Real-time controls in the right panel that adjust how data is summarized and displayed

All controls in the Plot Controls panel take effect immediately — plots refresh within milliseconds of any change. Your settings are automatically saved and restored on your next visit.

5.1 Summary Statistics

The **Summary stat** dropdown determines what the central tendency measure represents:

- **Mean:** Arithmetic mean of all values within the condition. Sensitive to outliers. Recommended for normally distributed data.
- **Median:** The middle value when data is sorted. Robust to outliers and skewed distributions. Recommended when data is non-normal.

5.2 Error Bars

The **Error bars** dropdown controls the vertical extent of error bars on the summary statistic:

Option	Full name	How it is calculated	Recommended when
SEM	Standard Error of the Mean	SD / √n (where n = number of replicates)	You want to show uncertainty of the mean estimate across replicate experiments
SD	Standard Deviation	Square root of variance across replicate means	You want to show the spread of replicate measurements
95% CI	95% Confidence Interval	Mean ± 1.96 × SEM	You want inferential interval bars that scale with sample size
None	No error bars	—	You prefer minimal clutter or are relying on the distribution shape alone

i ERROR BARS ARE COMPUTED ACROSS REPLICATE MEANS

Error statistics (SEM, SD, CI) are always computed across the per-file means, not across individual ROIs. This correctly treats each experiment as one biological replicate — the recommended approach for calcium imaging superplots.

5.3 Data Representation Checkboxes

Control	What it does
Paired lines	

	Draws connecting lines between per-file means across conditions. Use this when your experiments are paired — e.g., the same cell line measured under two different conditions on the same day.
Replicate dist.	Instead of showing all individual ROI values, shows only the distribution of per-file means. Useful when you have many ROIs per file and want to emphasize the replicate-level variability rather than ROI-level variability.
Log axis	Switches the Y axis to a logarithmic scale. Useful when comparing data that spans multiple orders of magnitude, such as event frequency.
Horizontal	Rotates the chart 90° so conditions are listed on the Y axis and values run left-to-right. Useful when condition names are long, or when you prefer a horizontal layout.
Grid	Shows or hides horizontal gridlines behind the plot. Uncheck for cleaner figures.
Conditions legend	Shows or hides the condition color legend in the Appearance panel.
Replicate legend	Shows or hides the replicate symbol/color legend in the Appearance panel.

5.4 Condition Ordering

The **Order** dropdown controls the left-to-right sequence of conditions on the X axis:

- **Entered:** Conditions appear in the order they were first assigned (default).
- **Manual:** You control the order yourself using ↑/↓ arrow buttons that appear next to each condition in the legend panel. Drag the order to match your preferred layout.
- **A–Z:** Alphabetical order by condition name.
- **By summary:** Sorted by ascending mean or median value — useful for ranking conditions.

5.5 Point Style Sliders

Slider	Range	Default	Effect
Point size	3 – 10	5	Radius of individual ROI scatter points. Increase for sparse data; decrease when points overlap heavily.
Point alpha	0.15 – 0.90	0.45	Opacity of individual ROI points. Lower alpha reveals density in overlapping regions (darker = more points). Higher alpha makes individual points more visible.
Jitter	0.08 – 0.32	0.18	Horizontal spread applied to scatter points to reduce overplotting. Increase if points stack on top of each other; decrease for a narrower, tidier appearance.

SECTION 6

Appearance Panel

Customizing colors, labels, fonts, and axis ranges for publication-ready figures

The Appearance panel is located below the Plot Controls in the right sidebar. All changes apply instantly to the active plot and are saved in the browser automatically.

6.1 Condition Colors

Color Palette

The **Palette** dropdown offers two preset color sequences:

DEFAULT PALETTE (12 COLORS)

A vibrant 12-color sequence optimized for general use on white or dark backgrounds.

OKABE-ITO PALETTE (8 COLORS)

A colorblind-safe palette (safe for protanopia, deuteranopia, and tritanopia). Recommended for figures intended for wide publication.

Per-Condition Color Override

Each condition in the legend has a color swatch and a hex color input. Click the swatch to open a color picker, or type a hex code directly. The ↺ **reset** button next to each condition restores its palette default.

Manual Condition Ordering

When **Order** is set to "Manual" (in the Controls panel), ↑ / ↓ arrow buttons appear next to each condition in the legend. Click them to move conditions up or down in the X-axis sequence. The order is saved automatically.

6.2 Labels & Title

Three text inputs allow you to override the automatically generated axis and title text:

Field	Default (Auto)	Example override
Plot title	Metric name + unit	"Peak $\Delta F/F_0$ — HEK293T cells"
X-axis label	"Condition"	"Treatment group"
Y-axis label	Metric name with unit	"$\Delta F/F_0$ (normalized)"

Leave any field blank to restore the automatic label. Click **Reset label overrides** to clear all three at once.

6.3 Typography

The Typography section lets you adjust font family and individual font sizes for every text element in the plot:

Setting	Default	Range
Font family	System Sans	System Sans / Arial / Helvetica Neue / Segoe UI / Georgia / Times New Roman
Plot title size	13 pt	8–32 pt
X-axis title size	11 pt	8–28 pt
Y-axis title size	11 pt	8–28 pt
X-axis tick label size	12 pt	8–24 pt
Y-axis tick label size	12 pt	8–24 pt
Legend font size	12 pt	8–24 pt
X-axis tick angle	Horizontal (0°)	0° / 45° / −45° / 90° (Vertical)

Click **Reset text** to restore all typography defaults at once.

6.4 Axis Ranges

The Axes section provides manual min/max inputs for both axes. Leave any field blank to use automatic (data-driven) scaling.

- **Y range** applies to all metric tabs.
- **X range** applies only to the trace tabs (Raw F and $\Delta F/F_0$) and controls the time window displayed.

Click **Reset ranges** to clear all custom axis limits and return to auto-scaling.

SECTION 7

Exporting Plots

Saving publication-quality figures using the Plotly toolbar

Every plot rendered by Ca^{2+}tchAll is an interactive **Plotly** chart. The Plotly toolbar appears in the top-right corner of each plot when you hover over it.

Toolbar button	What it does
📷 **Download plot as PNG**	Saves the current plot as a high-resolution PNG image file. The filename is auto-generated from the metric name. This is the primary export method for figures.
🔍 **Zoom**	Click and drag to zoom into a rectangular region of the plot.
✋ **Pan**	Click and drag to pan around the plot when zoomed in.
⛶ **Zoom in / Zoom out**	Step-zoom in or out by a fixed factor.
↩ **Reset axes**	Returns the view to the default zoom and pan position (does not affect the Axis Ranges settings).
Show/hide tooltips	Toggle data-point hover tooltips on or off.

💡 **TIPS FOR BETTER PNG EXPORT**

- Before exporting, use the Axis Ranges controls to set precise Y-axis min/max so all conditions are shown in the same frame across figures.
- Use the **Typography** section to increase font sizes if the exported image will be reduced to a small figure panel.
- Switch to the **Okabe-Ito** palette before exporting figures intended for colorblind readers.
- Enable the **Horizontal** layout in the Controls panel if condition names are long and would overlap on the X axis.

SECTION 8

Preferences & Persistence

What is saved automatically and how to manage your settings

Ca^{2+}tchAll saves the following settings to your browser's **localStorage** automatically. They persist across page refreshes and server restarts — nothing is lost when you close and reopen the app.

What is saved	Details
All plot controls	Summary stat, error bars, point size, alpha, jitter, all checkboxes, condition order
Chart style	Violin / Bar / Box / Strip / Raincloud
Active tab	The last metric tab you had open
Condition colors	Any manual color overrides per condition name
Color palette	Default or Okabe-Ito
Manual condition order	The sequence of conditions when using Manual ordering
Label overrides	Custom title, X label, Y label
Typography settings	Font family and all font sizes
Axis ranges	Custom X/Y min and max values
Theme	Dark or light mode
Condition assignments	The mapping of file names → condition names, restored when the same files are re-uploaded
Tour seen flag	Prevents the onboarding tour from auto-launching after the first visit

⚠ SESSION DATA IS NOT PERSISTED

The loaded files themselves (ROI values, trace data) are held in server memory only. They are lost when the server restarts or the browser tab is closed. You need to re-upload your `.xlsx` files each session. Condition assignments are restored automatically when you re-upload files with the same names.

SECTION 9

Onboarding Tour

The interactive 5-step guided introduction to the application

On your first visit, Ca^{2+}tchAll automatically launches a 5-step tour that highlights each main area of the interface. The tour can be skipped at any time and replayed later by clicking the **?** button in the bottom-left corner of the screen.

Step	Panel highlighted	What it explains
1 / 5	Experiments panel (left)	How to drag-and-drop or add files, and how to type condition names to group replicates
2 / 5	Tab bar	How metric tabs appear automatically once conditions are assigned
3 / 5	Controls panel	Summary stats, error bars, condition ordering, and the real-time update behavior
4 / 5	Appearance panel	Condition color customization, label overrides, fonts, axis ranges, and the Plotly toolbar
5 / 5	Theme toggle	Dark/light mode and the fact that all settings persist in the browser

Tour navigation:

- Click **Next →** or **← Back** to move between steps.
- Click **Skip** to dismiss the tour immediately.
- Click **Finish ✓** on the last step to complete the tour.
- Use `→` / `←` arrow keys to advance or go back.
- Press `Esc` to exit the tour at any time.

SECTION 10

Dark / Light Theme

Switching between light and dark mode

A theme toggle button is located in the top-right of the header bar. It reads **"Dark mode"** when you are currently in light mode (click to switch to dark), and **"Light mode"** when in dark mode (click to switch to light).

Dark mode applies across the entire interface — panels, plots, legends, and all UI controls adopt a dark color scheme automatically. Plot backgrounds, gridlines, and axis labels update to maintain readability.

Your theme preference is stored in the browser and restored on every visit.

> 💡 **NOTE ON EXPORTED FIGURES**
>
> The plot background color in exported PNG files matches the current theme. If you need white-background figures for publication, make sure to export in light mode.

SECTION 11

Keyboard Shortcuts

Available keyboard interactions

Keyboard shortcuts are currently limited to the onboarding tour:

Key	Context	Action
→ Arrow Right	Tour active	Advance to next tour step (or finish on the last step)
← Arrow Left	Tour active	Go back to previous tour step
Esc	Tour active	Exit the tour immediately

i PLOTLY CHART KEYBOARD INTERACTIONS

Plotly charts also respond to scroll wheel zoom (when hovering over a chart) and double-click to reset the view. These are built-in Plotly behaviors, not Ca^{2+}tchAll-specific shortcuts.

SECTION 12

Troubleshooting

Common issues and how to resolve them

Problem	Likely cause	Solution
No plots appear after assigning conditions	The condition name has only been typed for one file, or the Metrics sheet is missing from the uploaded files.	Make sure at least one condition has more than zero files with that condition name assigned, and verify the .xlsx file contains a "Metrics" sheet.
Upload fails with "bad_structure"	The file is not a valid Excel workbook, or it was exported from unsupported software.	Open the file in Excel, verify it has the expected sheet structure (Metrics, Raw_Traces, or DeltaF), and re-export from the Ca^{2+}tch-One.
Upload fails with "per-file RAM cap"	The file has too many ROIs or too many time points for the current session memory budget.	Remove some existing files to free up session RAM, or reduce the recording to fewer ROIs/ frames before exporting.
Trace tabs do not appear	Uploaded files lack Raw_Traces or DeltaF sheets, or the trace sheets exceeded the 500,000-cell limit and were skipped.	Check the file list for "traces skipped" warnings. If skipped, reduce the number of time points or ROIs in the export.
Condition auto-restore shows wrong names	The file was renamed since the last session, or the browser localStorage was cleared.	Re-assign condition names manually. New assignments will be saved for the next session.
Some metric tabs are missing	The loaded files do not contain those particular metrics in their Metrics sheet.	Tabs are hidden automatically for unavailable metrics. This is normal — a "Hidden metrics" note will appear in the controls area if metrics differ across files.
Server does not start	Python or required packages are not installed, or the port 5000 is in use by another application.	Check that Python 3.9+ is installed and run `pip install -r requirements.txt` in the backend folder. If the port is occupied, close the other application or edit the start script to use a different port.
Plot looks different after refreshing	Files are not loaded (session data is in-memory only) but	Re-upload your files. Condition names will be restored automatically from the previous session.

preferences are restored
from localStorage.

💡 **STARTING FRESH**

To reset all preferences and start with a clean slate: open your browser's Developer Tools, go to **Application → Storage → Local Storage**, and clear the entries for `localhost:5000`. This removes all saved settings including colors, typography, and condition assignments.